ICT. Ed. 516: Advanced Java Programming

Course no. : ICT. Ed. 516 | Nature of the course: Theoretical + Practical
Level: M.Ed. | Credit hours: 3 (2T+1P)
Semester: First | Teaching hours: 64 (32+32)

1. **Course Introduction**
   This course is a study on Java language techniques beyond the introductory coursewhich basically focuses GUI and event-driven programming, database connectivity, socket programming, distributed programming and servlets and JSP technology.

2. **Course Objectives**
   After the completion of this course, the students should be able to:

   - Introduce Basic Java Programming
   - Exemplify the concept of GUI programming and JDBC
   - Demonstrate socket programming. remote objects, and JSP technology

3. **Course Outlines:**

Specific Objectives	Contents	Teaching Hours
• Review of the object-oriented concept using JAVA language. • Demonstrate object-oriented concepts in including array, class, object, overloading, inheritance, interface package and files.	**Unit 1: Review of Programming Concepts in Java** 1.1. Java architecture, Java buzzwords, path and classpath variables, Sample Java program, compiling and running Java programs. 1.2. Arrays, for each loop, class and object, overloading, access privileges, interface, inner class, final and static modifiers, packages, inheritance, overriding. 1.3. Handling exceptions: Try, catch, finally, throws, and throw keywords, creating exception class 1.4. Concurrency: Introduction, thread states, writing multithreaded programs, thread properties, thread synchronization, thread priorities 1.5. Working with files: Byte Stream Classes, Character Stream Classes, Random Access File, Reading and Writing Objects. **Practical Work** • Array, class, object, overloading, inheritance, interface, package, files	12
• Describe the user interface in Java. • Handle the GUI control	**Unit 2: User Interface Components with Swing** 2.1. Introduction: Concept of AWT, AWT vs swing, Java applets, applet life cycle, swing class	12

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• Create a menu, toolbar and taskbar • Demonstrate GUIS components	hierarchy, component and containers 2.2. Layout management: No layout, flow layout, border layout, grid layout, gridbag layout 2.3. GUI Controls: Text fields, password fields, text areas, scroll pane, labels, check boxes, radio buttons, borders, combo boxes, sliders 2.4. Menu, menu item, icons in menu items, check box and radio buttons in menu items, pop-up menus, keyboard mnemonics and accelerators, enabling and disabling menu items, toolbars, tooltips 2.5. Option dialogs, creating dialogs, file choosers, color choosers, internal frames, frames. **Practical Work** • Components, containers, layout managers, menus, dialog boxes,	
• Demonstrate an event handling concepts in JAVA. • Demonstrate GUIS components	**Unit 3: Event Handling** 3.1. Event handling concept, listener interfaces, using action commands, adapter classes 3.2. Handling action events, key events, focus events, mouse event, window event, item events **Practical Work** • Listener interfaces, adapter classes	4
• Describe a database connection concept using JAVA and JDBC. • Demonstrate JDBC and DDL, DML statements.	**Unit 4: Database Connectivity** 4.1. JDBC architecture, JDBC driver types, JDBC configuration, managing connections, statements, result set, SQL exceptions 4.2. DDL and DML operations using Java, prepared statements, multiple results transactions, SQL escapes. **Practical Work** • JDBC steps, using DDL and DML statements	6

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• Explain the network programming using JAVA and network protocols. • Demonstrate socket programming with connection class.	**Unit 5: Network Programming** 5.1. Transmission control protocol (TCP), User datagram protocol (UDP), ports, IP address network classes in JDK 5.2. Socket programming using TCP, socket programming using UDP, working with URL's, working with URL connection class. **Practical Work** • Socket programming with TCP and UDP, URL and URL connection class	6
• Explain the GUI and JavaFX concept. • Demonstrate a layouts, control, menu, dialog box using JavaFX.	**Unit 6: GUI with JavaFX** 6.1. Introduction, JavaFX vs swing, JavaFX layouts: FlowPane, BorderPane, Hbox, VBox, GridPane 6.2. JavaFX UI Controls: Label, TextField, Button, RadioButton, CheckBox, hyperlink, menu, tooltip, FileChooser. **Practical Work** • Layouts, controls, menus, dialog box	4
• Explain the servlet and server pages • Demonstrate a servlet programing • Demonstrate a JDBC and ervlets. • Demonstrate a JDBC and JSP.	**Unit 7: Servlets and Java Server pages** 7.1. Web container, introduction to servlets, life cycle of servlets, the servlet APIs, writing srvlet programs, readingform parameters, processing forms, handling HTTP request and response (GET / POST request), database access with servlets, handling cookies 7.2. Servlet vs JSP, JSP access model, JSP syntax (directions, declarations, expression, scriplets, comments), JSP implicit objects, object scope, processing forms, database access with JSP.	14

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	Practical Work • Creating forms, processing forms, JDBC and servlets, JDBC and JSP	
• Describe the RMI and architecture. • Describe the CORBA and architecture. • Demonstrate the RMI program.	**Unit 8: RMI and CORBA** 8.1 Introduction to RMI, architecture of RMI, creating and executing RMI applications 8.2 Introduction to CORBA, RMI vs CORBA, architecture of CORBA, concept of IDL. **Practical Work** • RMI programs	6

9 Instructional Techniques

The instructional techniques for this course are divided into two groups. The first group consists of general instructional techniques applicable to most of the units. The second group consists of specific instructional techniques applicable to particular units.

4.1 General Techniques

Reading materials will be provided to students in each unit. Lecture, discussion, use of multi-media projector, brain storming will be used in all units.

4.2 Specific Instructional Techniques

Demonstration is an essential instructional technique for all units in this course the during teaching learning process. Specifically, demonstration with practical work will be a specific instructional technique in this course. The details of suggested instructional techniques are presented below

Laboratory Work: The students need to write programs related to basic java programming concepts, designing GUI, event handling, JDBC, network programming, web programming, and distributed programming.

5. Evaluation

Internal Assessment	External Practical Exam/Viva	Semester Examination	Total Marks
40 Points	20 Points	40 Points	100 Points

***Note**: Students must pass separately in the internal assessment, the external practical exam and the semester examination.*

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5.1 Internal Evaluation (40 Points)

The internal evaluation will be conducted by the subject teacher based on the following criteria.

11) Class attendance	5 points
12) Learning activities and class performance	5 points
13) First assignment (written assignment)	10 points
14) Second assignment (Case study/project work with presentation)	10 points
15) Terminal examination	10 Points
Total	40 points

5.2 Semester Examination (40 Points)

The Examination Division, Dean's Office, Faculty of Education will conduct the final examination at the end of the semester.

3) Objective questions (multiple choice 10 questions x 1mark)	10 Points
4) Subjective answer questions (6 questions with 2 'OR' x 5 marks)	30 Points
Total	40 points

5.3 External Practical Exam/Viva (20 Points)

The Examination Division, Dean's Office, Faculty of Educatuin will conduct the final practical examination at the end of the semester.

Practical Examination Evaluation Scheme (Total 20 Marks)

- Record book 4
- Laboratory work exam/case..... 8
- Viva voce..... 8

Total 20

10 Recommended books and Reference materials (including relevant published articles in national and international journals)

Recommended books

1. . Horstmann, C.S. (2018). *Core java volume I—Fundamentals (11th Edition)* Pearson,
2. Horstmann, C. S. (2019). *Core java volume II-Advance features,* Pearson.
3. Schildt, H. (2018). *Java: The complete reference* (11th Edition). McGraw-Hill Education.
1. D.T. Editorial Services (2015), Java 8 programming Black Book. Dreamtech Press.

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